

Application-aware DDoS Detection in High-Bandwidth Networks

Ruhollah Jahanafrooz

Abstract—High-bandwidth networks (HBN) which have high throughput and involve new-age protocols, are hard to manage. One of the serious challenges in these networks, is detection of Distributed Denial of Service (DDoS) attacks. High Velocity, Variety, and Volume (3V) are the three main features of High-Bandwidth networks. Over the past decades, many methods have been proposed to detect and mitigate DDoS attacks. Lack of adaptability to variety characteristic in high-bandwidth networks is the issue that is not mentioned in the previous works. In this research, we introduce an adaptive approach to the traffic variety in high-bandwidth networks for effective DDoS detection. This method classifies traffic flows to different layer 7 applications using a DPI tool, resulting in higher classification accuracy. It consists of three stages: Profiling, Anomaly Detection, and Mitigation. In the profiling stage, flow-level and packet-level statistical features are computed over time intervals to model the normal behavior of each application protocol. Anomalies are then detected using a simple yet effective statistical technique. Upon detection, the system identifies anomalous groups of users associated with the targeted application protocol and blocks their corresponding flows to mitigate the attack. **Experimental results show a ??% average accuracy in identifying application layer DDoS attacks of different datasets. These results indicate our novel approach can lead to developing more efficient, low FPR methods to detect and mitigate modern DDoS attacks.**

Index Terms—DDoS attacks, High-Bandwidth Networks, Application Protocols, Intrusion Detection Systems.

1 INTRODUCTION

Denial of Service (DoS) attacks aim to disrupt a server's ability to respond to legitimate user requests, preventing it from delivering its intended services under normal conditions. Distributed Denial of Service (DDoS) attacks are more complex types of these attacks and can generate more traffic rate compared to simple dos attacks. Also it is more challenging to identify the source of DDoS attack and mitigate them. Today, these attacks can be executed easily with the help of increasing number of vulnerable Internet-of-Things (IoT) devices and free available DDoS tools. The attacker is able to take control of these vulnerable devices easily and use them as a bot to flood the victim server. **This explains the reason behind the growing number of attacks in recent years [1]. According to the latest Cloudflare report, they mitigated 14 million attacks in 2023.** They also reported the mitigation of 8.5 million attacks until the second quarter of 2024, which increased by 20% to the previous year. Mirai botnet (a special botnet that infects many IoT device) triggered the largest number of DDoS attacks since 2016. The data transmission rate can exceed 600 Gbps. Mirai botnet was used in ddos attacks against many popular websites such as Etsy, Netflix, Shopify, SoundCloud, and Twitter, by sending traffic to DNS providers. In March 2018, GitHub suffered from the most serious DDoS attack ever with 1.35 Tbps peak traffic [2]. In 2022 Microsoft reported that it has mitigated near to 1000 attacks in a day. 2125, is the most number of attacks in a day reported also by Microsoft [3]. By Akami report, more recently, Anonymous Sudan claimed responsibility for the attack on France's Telegram messaging app as part of the unprecedented DDoS attack on the country's state inter-ministerial network, causing the disruption of more than 17000 IP addresses and devices and more than 300 domains.

Application Layer attacks are a type of ddos attacks that mimics the benign users' traffic but with the goal of

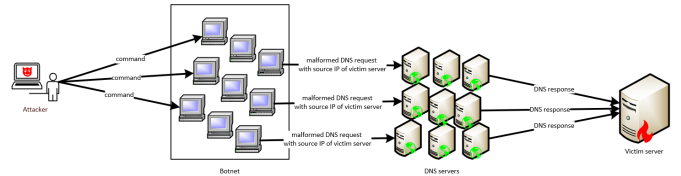


Fig. 1. Reflective DDoS attack using DNS service

directing enormous number of requests to victim server. These types of attacks are typically carried out using vulnerable reflective servers, as illustrated in Figure 1. As for an application server, processing a user's request requires more resources than creating it. This difference is known as the Amplification factor. Amplification factors for some application layer protocols are depicted in Figure 2. According to a published record by Akami, application layer DDoS attacks are evolving faster and replacing traditional DDoS attacks. However, network-layer attacks still persist, as they are simpler and less resource-intensive to execute. [4]. In 2022, an HTTP based attack with 4.6 billion packets per second, targeted Google Cloud Armor [5].

Type	Port	Amplification Factor
NTP	UDP 123	556.9
CLDAP	UDP 389	56 – 70
TFTP	UDP 69	60
DNS	UDP 53	28 – 54
SNMP	UDP 161	6.3
MSSQL	UDP 1434	25
NETBIOS	UDP 137	3.8

Fig. 2. Application layer protocols and their amplification factor

Application-layer attacks exploit the mechanisms of client-server communication, meaning the attacker must possess a thorough understanding of protocol behavior and communication infrastructure [6], [7]. As mentioned earlier, these attacks closely mimic legitimate user traffic patterns, making their detection significantly more challenging. Consequently, static signature-based detection methods are no longer effective [7]. However, even in such attacks, the malicious traffic often exhibits distinctive statistical characteristics at both the flow-level and packet-level compared to normal traffic [8].

Most previous methods have focused on detecting only specific types of DDoS attacks. To achieve comprehensive coverage of various DDoS attacks, one might consider deploying multiple detection and mitigation mechanisms simultaneously; however, this approach is impractical due to the complexity and overhead of managing numerous systems. Therefore, a solution capable of dynamically adapting to different traffic profiles is required. In this paper, we present a DDoS mitigation framework designed for high-bandwidth networks, named ADD (Application-level DDoS Defense). Our approach relies only on statistical methods, which eliminates the need for infrastructure modifications. Compared with the previous methods, the main contributions of this work are as follows:

- 1) Application-aware detection: By profiling each application protocol using statistical features of its normal behavior, anomalous activities specific to that application protocol can be effectively detected. Since there are considerable variations across different application protocols, it is essential to maintain separate profiles for each one [9]. This approach ultimately enhances detection accuracy and also facilitates mitigation process.
- 2) Detection and mitigation of various Layer 7 ddos attacks: Our method can accurately identify both the victim application protocol and the attacking sources, enabling effective and targeted mitigation. Unlike most previous approaches, which focus solely on detection, our solution provides an integrated mechanism for both detection and mitigation
- 3) Cost efficiency: Since our approach relies on lightweight statistical analysis, it imposes minimal overhead on network infrastructure. Also, in contrast to machine learning-based methods, it does not require extensive training data or computational resources.

The rest of this paper is organized as follows: An overview of recent defense methods in the high-bandwidth networks are given in Section 2. The adaptive detection method for high-bandwidth networks is introduced in Section 3. Empirical simulation and evaluation of ADD is explained in Section 4. Finally, we highlight the ongoing challenges and discuss the future direction in Section 5.

2 RELATED WORK

Recent DDoS detection methods can generally be classified into two main categories: signature-based and anomaly-based approaches. Signature-based methods rely on predefined patterns or known attack signatures to identify

malicious traffic. In contrast, anomaly-based methods construct a behavioral model of normal traffic using selected features and detect deviations that exceed a predefined threshold, which is typically determined during the training phase. Furthermore, anomaly-based detection techniques can be subdivided into two groups: statistical-based methods and machine learning-based methods. Although signature-based approaches offer faster detection due to lower computational overhead, they are ineffective against novel attack patterns. The main challenges faced by these methods in detecting DDoS attacks in high-bandwidth networks, are as follows:

- 1) **Defining the normal state according to the extent and variety of the traffic:** Since the incoming network traffic includes data from multiple applications, defining an accurate model of normal traffic behavior is challenging. To create an effective model, numerous traffic features can be extracted and used as discriminative characteristics. If these features are not selected appropriately, the accuracy of the method may become suboptimal.
- 2) **Real time alerting of attacks:** The massive volume and continuous flow of network traffic, combined with the requirement to notify users of detected anomalies in real time, make anomaly detection both challenging and critical. Due to the high volume of recent DDoS attacks, detecting and mitigating them as quickly as possible, can enhance the Quality of Experience (QoE)¹. Real time detection, can also prevent disruptions in network performance and maintain the overall Quality of Service (QoS). This challenge is especially evident in the machine learning-based detection methods, which will be discussed in the following section.

2.1 Statistical Anomaly-based Models

In 2018, Kalkan [10] developed a method that whenever a congestion occurs, identifies the most suspicious attribute pairs and mitigates the attack by blocking packets matching those attribute combinations. However, the method cannot detect slow-rate attacks, and the threshold used for suspicious-pair detection is not clearly specified. The mitigation strategy blocks packets with specific attribute values (for example: TTL=10, DstPort=90), which can inadvertently block legitimate traffic and thus yields a high false positive rate (FPR). The authors also noted that network profiles may vary across days and times of day, which can further reduce effectiveness of the proposed method. But we argue that we overcome this challenge by using application aware DDoS detection method.

One of the most significant characteristics of Denial of Service attacks is sending a large number of packets. These packets all use the same protocol with a small payload size, between 40 and 60 kilobytes, which also have the same destination addresses but random destination ports. In 2019, David et al [11] presented a solution that uses a simple statistical detection method. This method measures

1. It reflects the satisfaction of legitimate users while using network services

the number of packets, the number of unique source, the destination IP addresses, as well as the number of unique protocols. It obtains the value of these features during each time window and calculates the dynamic threshold using the variance and average parameters. In this method, only the improvement of detection accuracy and the reduction of false negative rate is addressed. Since the method was evaluated using the old DARPA dataset, which was prepared in 1998, its performance in today's networks may not be reliable. This dataset does not include today's application layer attacks. For this reason, the method is not guaranteed to detect various attacks in high bandwidth networks.

In 2021, Shi et al. [8] proposed RT-SAD, which detects potential attacks by analyzing the imbalance of flow attributes across destination addresses. The method uses Count-Min (CM) sketches, compact probabilistic data structures that efficiently summarize traffic statistics while reducing memory usage. For each destination IP, RT-SAD calculates thresholds based on the mean and variance of the observed flows. Although effective in detecting attacks, the method cannot identify the victim's address and lacks a systematic approach for selecting features to cover diverse attack types.

Hagimaghsoodi et al. [12] also presented a reactive solution in 2022. The Proposed method has three phases of detection and defense mechanisms for DDoS attacks. In the first phase, the users are ranked based on their traffic behaviour. The second phase identify potential ddos attacks by observing unusual behaviour in network status. If an anomaly is detected, the third phase will enforce restriction policies on the most suspicious users according to their accumulated ranking in the first phase. However, the method is difficult to deploy in practice and doesn't clearly describe the detection process.

Shao et al. [13] proposed a solution in 2023 that employs a specialized sketch structure to store statistical traffic features, enabling high detection accuracy even when traffic is sampled. This method integrates multiple pre-trained classifiers, each optimized for specific attack types and behaviors. This fine-grained classifiers make it able to achieve fine-grained detection which addresses the limitations of prior methods such as RT-SAD in identifying attacker fingerprints (e.g., victim or exploited protocol). However, the approach is limited to detecting UDP-based flooding attacks and performs poorly against previously unseen attack patterns. Moreover, it does not provide a mechanism for online model re-training.

In 2024, Zaho et al. [14] proposed a method that can detect 15 different types of attacks with only a single flow rule. This achieved by representing the relationships between them as graphs, followed by partitioning them into sub-communities with specific algorithms. As authors mentioned, the computational complexity may rise with more categories, which could hinder real-time processing in high-bandwidth environments where rapid response is critical.

2.2 Machine Learning Anomaly-based Models

In recent years, machine learning and deep learning-based methods gained considerable attention as an efficient method to enhance network security. These methods

are also used in the field of detecting distributed denial of service attacks. Different types of learning algorithms, including classical machine learning algorithms (such as support vector machine, Naïve Bayesian, decision tree, etc), neural networks and deep neural networks, self-organizing mapping, and so on, are used to perform flow classification. However, high computational overhead compared to statistical methods has made the use of these methods not so suitable for real-time attack detection. Furthermore, in supervised machine learning methods the need to use a well-labeled dataset for training the model, is a serious challenge.

In 2021, Dimolianis et al. [15] proposed a method that generates attack signatures which are extracted from passing network packets using Deep Packet Inspection (DPI). The approach first extracts statistical characteristics from the traffic and then classifies them as normal or abnormal signature using a supervised machine learning model. Then, based on the detected attack type, it automatically generates filtering policies to block malicious traffic and deploys these rules on programmable switches. However, this method is inefficient in identifying attacks with varying traffic patterns and does not address the challenge of retraining classification models or managing the high computational cost associated with real-time operation in high-bandwidth networks.

Wang et al [16] proposed a method in 2022 for detecting and mitigating DDoS attacks in SDN. This method consists of an anomaly detection module using a hybrid CNN-ELM model and a mitigation module employing IP traceback technique. However, the limitation is its reliance on supervised learning methods, which require extensive labeled data, making it costly and impractical for real-world applications.

Myneni et al [17] in 2022, presented a method called Smart-Defense. This method, mitigates malicious traffic by deploying neural network-based identification modules on both the source and destination sides. The system also includes a mechanism bot detection. However, the requirement to implement neural networks on both the source and destination side is one of the challenges of implementing and using smart-Defense in real world scenarios. Also, not providing an optimal solution for retraining the model in a fast way, is another disadvantage of this method.

In 2023, Xie et al. [18] presented a solution to better distinguish normal user traffic from attack traffic in the application layer. This method tries to obtain a communication model of application layer protocols. To achieve this, it extracts keywords of each application layer protocol using predefined regular expressions. It gathers the time series of these observed keyword in time intervals. Finally, it sends this sequence as input to a recurrent neural network that has been trained to detect attacks. The problem of this method is the dependence of its neural network-based model on application layer protocol's keywords. These keywords must be explicitly defined beforehand. This means that the proposed solution cannot effectively detect attacks performed in newer application layer protocols. In addition, the time-consuming process of updating the model in diverse high-bandwidth traffic also remains unsolved.

In 2024, Lie et al. [19] designed a framework for DDoS

detection in large scale networks. They used a novel algorithm to effectively reduce dimensionality and background noise in large-scale network data which involves numerous traffic features. It utilizes tensor space smoothing and reconstructing for preprocessing, followed by XGBoost classification for DDoS attack detection. However, the evaluation was conducted on an outdated dataset, which limits the method's applicability in real-world scenarios.

Yu et al. [20] in 2024, proposed a method that combines Rényi entropy with a fixed time window and an improved Exponentially Weighted Moving Average (EWMA) model to create a dynamic threshold for anomaly detection. The approach first selects key network traffic features and computes their Rényi entropy. Using the improved EWMA model, it predicts the entropy in the next time window and generates an adaptive threshold to identify abnormal traffic. On the CICDDoS2019 dataset, the method achieved an F1-score of 0.927% for the DoS attack category. However, it cannot identify the attacker.

Hirsi et al. [21] in 2025, proposed a hybrid model of Random Forest for feature selection and SVM for traffic classification. This method uses a dynamic user trust score to block malicious sources and mitigate attacks. However, the study does not describe how the model adapts to new traffic patterns. Also the authors acknowledge the high computational complexity of their approach when applied in large-scale environments.

Identifying denial of service attacks has long been one of the most significant challenges for network security researchers. Previous methods mainly utilized statistical analysis or pre-trained machine learning models. Statistical methods are generally faster than machine learning methods due to their low computational overhead. Machine learning methods can achieve high detection accuracy if properly implemented and regularly updated with sufficient datasets. In this section, recent studies on detecting DDoS attacks in high-bandwidth networks were reviewed. We also mentioned the limitations of each one. A summary of the methods reviewed in this section is provided in table 1. This table outlines the characteristics of each method along with their strengths and weaknesses in detecting DDoS attacks in high-bandwidth networks. Failing to consider the high traffic variety in high-bandwidth networks is a common limitation among all these approaches.

TABLE 1
Previous methods

Method	Mechanism or Measures		High-bandwidth Networks		
	Detection	Mitigation	high Velocity	high Volume	high Variety
Kalkan 2018	×	✓	×	×	×
David 2019	✓	×	×	✓	×
Shi 2021	✓	×	✓	✓	×
Hagimaghsoodi 2022	×	✓	×	×	×
Shao 2023	✓	✓	✓	✓	-
Zhao 2024	✓	✓	-	✓	✓
Dimolanus 2021	✓	✓	-	✓	×
Wang 2022	✓	×	✓	✓	×
Myneni 2022	✓	✓	×	✓	×
Xie 2023	✓	✓	×	✓	×
Yu 2024	✓	×	-	-	-
Hirsi 2025	✓	✓	-	-	-

3 ADAPTIVE DDoS DETECTION

High-Bandwidth networks are characterized by 3 main criteria also known as the 3V model:

- 1) High Velocity: Network traffic is being generated and transmitted at a high speed. In the recent years with the advent of Ethernet100 and Cisco Infini-band, network devices can handle more than 100Gb/s rate of data [6]. Also in 5G, each user is capable of sending 400 MB per second on average [22].
- 2) High Volume: Data passing through the network, consists of huge metadata including an enormous number of payloads and protocol headers. Headers are crucial for encryption, routing and other protocol requirements. However, managing and analyzing such a huge amount of data is very expensive. In 2003, the global network traffic volume was approximately 5 exabytes, and it reached 14.7 exabytes in 2008, representing almost a threefold growth. **It is estimated that by 2025, annual network traffic will reach 175 zettabytes** [23].
- 3) High Variety: There are huge number of protocols in application layer. Each application may employ its own specific protocol. Also it is possible to use different protocols while a single user is interacting with a service. In addition to the diversity of protocols, we have another challenge. Many applications may use a generic protocol, but each application can have a specific behaviour and interpretation of this protocol. For example in today's networks, most traffic is based on HTTPs passing through browsers, but the content of each packet is different for each application [24].

As discussed earlier, traffic variety is a topic that has not been deeply explored in previous researches. By traffic variety we refer two main concepts. First, it denotes the large number of domain-specific application-layer protocols used by different services. According to the **Applopedia Database²**, **the number of such protocols exceeded 4,000 by 2024**. Each of these protocols differs in terms of header structures, packet sizes, and, most importantly, their **client-server communication mechanisms**. Second, even when two applications use the same protocol, their traffic behavior is not identical. For example, many applications use the HTTP/s protocol to communicate with their APIs and clients, yet each application can produce a different traffic fingerprint. To further illustrate this, we examined five different web applications that all use HTTP/S as their layer-7 protocol for communication and data transmission. We captured the daily traffic of a normal Internet user while using these applications and extracted various traffic features, as shown in table 2. Similar research also has done in other studies which is depicted in figure 3.

As shown in table 2, for profiling network traffic with better accuracy, we need to account differences between applications. Each application exhibits a distinct traffic behavior, which makes it impractical to define a single static threshold for all of them. In other words, traffic must first be

2. <https://applipedia.paloaltonetworks.com>

TABLE 2
Different Web Applications Traffic Features

Application	Category	Avg Pkt Size (bytes)	Throughput (bits/sec)	Sessions (TCP)	Bytes
Youtu.tv	Video On Demand Streaming	633	20×10^3	12	819607
Google Maps	Navigaton Service	843	40×10^3	15	1687609
Gmail	Email Service	460	2.766×10^3	6	96635
Tumblr	Blog, Social Network	1130	7788×10^3	9	317170267
LinkedIn	Business Social Media Platform	690	105×10^3	3	4062742

	Service	Measured					Theoretical
		Instant messaging	Web (small objects)	Web (large objects)	Video streaming	App download	
	Provider	WhatsApp	Freeipk	Vimeo	YouTube	Google Play Store	Full buffer
Transport	Protocol	TCP	TCP	TCP	UDP	TCP	—
	Header [bytes]	32	32	32	8	32	32
	Max. DL payload [bytes]	147	1348	1348	1350	1348	1348
	Avg. DL packet length [bytes]	71	1139	1396	1189	1391	1348
	Max. DL packet length [bytes]	179	1380	1380	1358	1380	1380
	DL packets with MSS [%]	0	73	99	86	99	100
	No. of DL packets	27	56	2369	1988	30754	N_p
	No. of UL packets	30	39	1156	313	10991	N_p
	Ratio DL/UL packets	0.90	1.44	2.05	6.35	2.80	1
	Ratio UL/UL packets	0.90	1.44	2.05	6.35	2.80	1
IP	Protocol header [bytes]	20	20	20	20	20	20
	Max. packet length [bytes]	199	1400	1400	1378	1400	1400
PDCP	Total DL volume [kB]	1.9	63.8	3306.2	2362.9	42770.9	$1400 \times N_p$
	Total UL volume [kB]	2.5	4.9	61.9	80.08	571.5	$52 \times N_p$
	η_{UL} [%]	56.6	7.2	1.8	3.2	1.3	3.58

Fig. 3. Different applications fingerprints

classified, and then the behavior of each application should be profiled separately. Traffic classification solutions can generally be categorized into the following groups based on the underlying method [25]:

- 1) TCP/UDP ports: This approach identifies traffic based on well-known application ports. However, it can only detect predefined ports and can easily be bypassed.
- 2) QOS labels: These methods rely on specific fields set in the IP header to prioritize certain types of traffic [26]. Nonetheless, these labels can also be manipulated or bypassed.
- 3) Statistical features: These methods use statistical characteristics of packets (such as volume, flags, source/destination IPs, port numbers) and flows (such as flow duration and frequency). However, they are often ineffective for today's complex and encrypted traffic.
- 4) Deep Packet inspection: These methods inspect the packet payload to achieve more accurate classification, but they introduce significant computational and memory overhead.

nDPI was introduced in 2012 under the GNU license as an open-source tool for Deep Packet Inspection (DPI). It can detect more than 300 application layer protocols. It classifies network traffic into 20 categories, including media/audio, VPN, email, file sharing, social networks, and videogames. In addition, nDPI allows users to define custom protocols through configurable protocol definition file [27].

3.1 Proposed method

We propose a proactive, multi-attribute, application-layer DDoS detection method named ADD(Adaptive DDoS Detection). Since detecting an ongoing attack before it occurs is impossible, our approach focuses on building nominal traffic profiles for each application based on user-level,

flow-level, and packet-level features. This enables accurate differentiation between normal and malicious behavior. The ability to detect both the targeted application and the sources of attack allows the proposed method to implement targeted mitigation. Previous methods commonly assume that certain traffic characteristics remain inherently stable during normal network operations. However, this assumption does not hold in practice, as each application exhibits distinct traffic patterns. For instance, traffic passing through an Internet Service Provider (ISP) — a typical example of a high-bandwidth network — consists of diverse application services, each with its own protocol and behavioral characteristics. Today we have a lot of different services and each has its own protocol and traffic behaviour.

There are many statistical methods to model traffic and detect anomaly, which some of them are: WMA, CUSUM, ARIMA, Shewhart [28]. We use Exponential Weighted Moving Average (EWMA) method to detect anomaly in each application.

Previous studies have typically profiled network traffic based on packet-level or flow-level features. However, these approaches tend to suffer from reduced accuracy and higher false-positive rates, particularly in high-bandwidth networks. However, the RAD framework [12] proposed a user-based profiling strategy, shifting the focus from packet or flow characteristics to behavioural attributes of users. Building upon this idea, we introduce a novel profiling approach that models network behaviour using the $\langle user, application \rangle$ pair. To the best of our knowledge, this method has not been explored in prior works. Our profiling operates at the application layer, which is conceptually above both the packet and flow levels. When models are constructed based on packet granularity and mitigation thresholds are derived from packet features, legitimate packets from control flows can inadvertently be dropped—especially since some control flows may consist of only a single packet. Similarly, flow-level profiling lacks sufficient precision due to the diversity of application behaviours in high-bandwidth environments. An overview of our proposed method, along with the interaction among its main components, is illustrated in Figure 4.

Before delving into the details of ADD, we first provide a high-level overview of our proposed method:

- Profiling (Pre-processing): In the profiling phase, incoming packets are first aggregated into flows and labeled based on nDPI classification results. For each time interval, we record and profile the traffic behaviour of every identified application to establish its normal behavioral profile.
- Detection: According to Imperva's report [29], over one-third of DDoS attacks in the first half of 2021 lasted only one to four minutes, highlighting the critical importance of rapid detection and response. Traditional machine learning or deep neural network models, while accurate, often introduce considerable computational overhead, making them unsuitable for real-time DDoS mitigation in high-speed networks. Therefore, we adopt a lightweight statistical anomaly detection method, which avoids such overheads. If an anomaly is detected in an application,

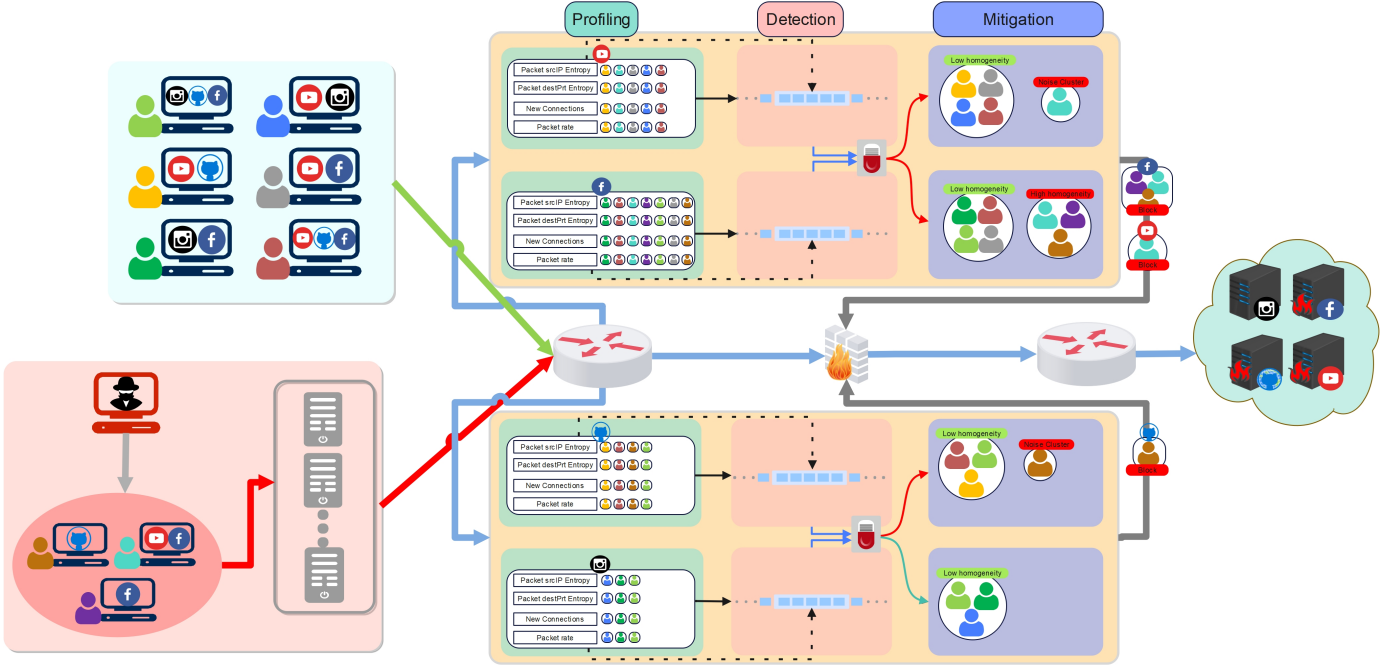


Fig. 4. Method architecture

TABLE 3
Terms and Parameters

Term	Explanation
pkt	The set contains extracted necessary information for each packet
fl	A set contains all available flows
flow(p)	The flow which packet p belongs
PSE(a)	The packet size entropy of the packets received by application protocol a
SIE(a)	The source IP entropy of the packets received by application protocol a
PPS(a)	The number of packets received by application protocol a
NS(a)	The number of SYN packets received by application protocol a
TS(u,a)	The number of active sessions of application protocol a initiated by user u
APS(u,a)	The average packet size of application protocol a sent by user u

the system immediately alerts and proceeds to the mitigation phase.

- **Mitigation:** We cluster users of each application according to their traffic behaviour, base on 3 features. When an anomaly is reported for a specific application, we analyze the user clusters of the application and the noise clusters are considered as malicious user(s).

We consider time interval T as 1 second. The incoming traffic behavioural pattern is calculated at the end of each time interval. In the following sections, we will fully describe 3 modules of our method. The parameters and relevant terms in the first phase are described in Table 3 with details explained in the following subsections:

3.1.1 Profiling

When a DDoS attack occurs, certain traffic features exhibit anomalous behaviour. In an ISP-level environment, where

a vast volume of packets and flows—each with numerous features—passes through the network, it is essential to select only the most discriminative features. Retaining all features would impose excessive computational and storage overhead. The profiling module captures and models the behaviour of different applications using these selected traffic features, establishing a baseline and corresponding threshold for each. **Note that all the following calculations are performed within a time interval of length $T(t) - T(t - 1)$.**

- **Packet Labeling:** We define a packet structure to describe typical packet information:

$pkt : <SourceIP, SourcePort, DestinationIP, DestinationPort, TransportLayerProtocol, PacketSize, TCPFlags, TimeStamp > .$

pkt consists the set of extracted necessary information for each packet.

We define a flow as an aggregation of packets. A flow is defined by five attributes as below:

$fl : <SourceIP, SourcePort, DestinationIP, DestinationPort, TransportLayerProtocol, ApplicationLabel > .$

Each flow has a application label that determines which application protocol it belongs. We use nDPI to determine this label.

Each packet belongs to a flow. Packets belong to a

flow if the first five attributes had the same value:

$$\begin{aligned} flow(p) : \{f | &SourceIP(p) = SourceIP(f), \\ &SourcePort(p) = SourcePort(f), \\ &DestinationIP(p) = DestinationIP(f) \\ &, DestinationPort(p) = \\ &DestinationPort(f), \\ &TransportLayerProtocol(p) = \\ &TransportLayerProtocol(f)\}. \end{aligned}$$

- **Application Profiling:** After classifying packets, we calculate below features of each application at the end of every time interval:

Packet size entropy (PSE): As depicted in table 2, different applications follow a certain packet size distribution. So whenever the entropy of packet sizes changes, it means that some abnormal packets are being sent to victim.

Entropy calculation works by analysing the frequency of distribution and calculating the probability of each value. We use Shannon Entropy which is calculated as below for a probability $p(x)$ [30]:

$$H(x) = - \sum_{i=1}^n p(x_i) \log_2(P(x_i))$$

PSE for application protocol a is calculated as below:

$$\begin{aligned} PSE(a) &= - \sum_{i=0}^n p(xps = i) \log_2(p(xps = i)) \\ p(xps) &= \frac{|xps|}{|total_packets|} \\ xps &= \{PacketSize(x) | x \in pkt, \\ &ApplicationLabel(flow(x)) = a, \\ &T(t-1) \leq Timestamp(x) \leq T(t)\} \\ total_packets(a) &= \{pk | pk \in pkt, T-1 \leq \\ &Timestamp(pk) \leq T\} \\ n &= 61536(\text{Available packet sizes}) \end{aligned}$$

Source IP entropy (SIE): In certain DDoS attacks, a sudden increase in the number of sending hosts (zombies) targeting the victim can serve as an indicator of anomalous activity. SIE(a) is calculated as follows:

$$\begin{aligned} SIE(a) &= - \sum_{i=0}^n p(xsi = i) \log_2(p(xsi = i)) \\ p(xsi) &= \frac{|xsi|}{|total_packets|} \\ xsi &= \{sourceIP(x) | x \in pkt \\ &flowlabel(flow(x)) = a, \\ &T(t-1) \leq Timestamp(x) \leq T(t)\} \\ n &= \text{Total number of unique IPs} \end{aligned}$$

Packets per seconds (PPS)(Traffic rate): pps is calculated as the number of packets received within each time interval. It is an important indicator that can

assist in identifying potential DDoS attacks. PPS is formally illustrated follows:

$$PPS(a) = \frac{|total_packets(a)|}{T(t) - T(t-1)}$$

New Sessions (NS): If a packet SYN flag was set to 1, we consider it as a new session. NS is calculated as follows:

$$\begin{aligned} NS(a) &= |xns| \\ xns &= \{pk | pk \in pkt, \\ &ApplicationLabel(flow(pk)) = a, \\ &"SYN" \in TCPFlags(pk), T(t-1) \leq \\ &TimeStamp(x) \leq T(t)\} \end{aligned}$$

We calculate each of these features for each application in every time interval T . Algorithm 1 shows how the profiling procedure.

Algorithm 1 Application Profiling

Input: $V[m] \leftarrow$ statistical feature value for each Application, $n \leftarrow$ current interval, $w \leftarrow$ sliding window size, $p \leftarrow$ profiling periods

for p **do**

calculate V_{max} and V_{min} for each feature of different application protocol

end for

for each time interval t **do**

update statistical feature: V_{t+1}^{actual}

Calculate WMA of current window: $V_{t+1}^{predict} = \sum_{i=t-w+1}^t \lambda V_{t+1}^{actual}, \sum_{i=t-w+1}^t \lambda = 1$

end for

3.1.2 Detection

The rate of change in traffic features differs significantly between normal and attack traffic. The key idea behind DDoS detection is that, as an attack progresses, the rate of change increases sharply. So the challenge is to identify the anomaly in the incoming traffic input. A series of the methods for detecting anomaly in general behavior of the data set is called moving average algorithms. These algorithms obtain the average of the data set in certain time intervals and calculate the average profile which results in removing noise and possible errors and gives more expression to the prominent traffic characteristics. We compare current interval of each application profile with defined threshold and alert if we encounter an anomaly bypassing a defined threshold. The detection algorithm is described in algorithm 2. As discussed in the profiling phase, we calculate the normal traffic behaviour of each application to predicate its expected traffic behaviour. The threshold is defined using the 3-sigma criterion based on the normal distribution, which enables us to identify anomalies as values that deviate significantly from historical records. The value that breaks the normally distributed values is considered as outliers or anomalies. This approach effectively detects deviations between the current traffic behavior and the established baseline.

Exponential Weighted Moving Average (EWMA): The EWMA provides a method to compute the average value for a set of data points (traffic flows) where the more recent data are given greater weight than the older data, ensuring that a dynamic threshold is applied based on overall system behaviour to detect irregular behaviour by examining the trends in traffic behaviour [31]. The principles of the EWMA model are illustrated in :

$$\bar{y}_t = \begin{cases} \alpha y_{t-1} + (1 - \alpha)\bar{y}_{t-1} & , t > 2. \\ y_1 & , t = 2. \end{cases}$$

Recording all observed values of traffic starting from the network is excessively complex, and the longer the duration of traffic generation, the smaller its impact on the results. Therefore, we establish a sliding window with size w , representing that the EWMA model will utilize historically observed values within the most recent w time units to predict the current value. In network traffic data, t represents the sequence of traffic data, and the preceding w sliding windows denote the extraction of the most recent w traffic data points from the dataset. The input values for the EWMA model are the statistical feature values, denoted as y , for each application. $\bar{y}_t$ is the predicted statistical feature value at t , $\bar{y}_{t-1}$ is the predicted statistical feature value at $t - 1$, and y_{t-1} is the actual statistical feature value. This is illustrated in bellow:

$$\bar{y}_t = \alpha y_{t-1} + \alpha(1 - \alpha)y_{t-2} + \dots + \alpha(1 - \alpha)^{w-1}y_{t-w}$$

The sliding window signifies the utilization of only the most recent w historical values as active data. After each prediction, the sliding window moves forward to incorporate the latest observed data while excluding the oldest data.

3-sigma Rule: We use 3-sigma normal distribution algorithm for defining threshold and detecting anomaly. According to the central limit theorem, a set of independent random variables, each with a finite mean and variance, tends to follow a distribution that approximates a normal distribution as the number of variables increases. For instance, although several factors can influence the measurement error of a quantity—such as instrument error, reading error, or environmental conditions—the distribution of errors in repeated measurements typically follows a normal distribution, with values scattered around a central mean. The slope of the probability distribution curve increases near the mean, and in practice, approximately 73.99% of the data lies within three standard deviations of the mean. This property, known as the 3-sigma rule or the empirical rule of the normal distribution, provides a useful guideline for understanding the spread and variability of data [32].

These methods can allow us to detect flood and high speed DDoS detection as well as low rate DDoS detection.

3.1.3 Mitigation

Whenever current interval has anomaly, we can easily detect the app and block $\langle user, app \rangle$ of that application that has the surpass the threshold User. For the mitigation, we do the following steps.

- User Profiling: We record each user of victim application in time interval T . Following features will be calculated for each user:

Algorithm 2 Anomaly Detection

Input: $V[m] \leftarrow$ statistical feature value for each Application, $t \leftarrow$ current interval, $w \leftarrow$ sliding window size
output: Attack alert

Calculate ratio metric $\leftarrow R_{n+1}^{predict} = \frac{V_{n+1}^{actual}}{V_{n+1}^{predict}}$

Calculate the mean and standard deviation for ratio metric mean value $\leftarrow R_{mean}$ and variance $\leftarrow \sigma$ of this ratio
 Use this method to calculate allowed margin threshold

$R_u \leftarrow R_{mean} + 3 * \sigma$

$R_l \leftarrow R_{mean} - 3 * \sigma$

if ($R_{t+1}^{predict} > R_u \& V_{t+1}^{actual} > V_t^{max}$) || ($R_{t+1}^{predict} < R_l \& V_{t+1}^{actual} < V_t^{min}$) **then**

DDoS alert!

else

Normal state, no attack occurred.

ratio R_{t+1} and statistical feature V_{t+1}^{actual} to window list $R[w]$ and $V[w]$ added correspond.

end if

Active Sessions: The number of active sessions for a user u of application protocol a can formally defined as below:

$$TS(u, a) = |x_{ts}|$$

$$x_{ts} = \{f | f \in fl, flowlabel(f) = a, \\ \forall pk : flow(pk) = f, \\ min_timestamp(f) - T(t) \\ < 300 \\ , "RESET, FIN" \notin TCPFlags \\ (pk)\}$$

$$min_timestamp$$

$$(f) =$$

$$\{min(TimeStamp(pk)) | \\ pk \in pkt, flow(pk) = f\}$$

A flow is inactive if either it has been terminated (through FIN or RESET) or it has not received any packet for 300 seconds.

Average Packet Size: The average packet size for a user u of application protocol a is defined formally as below:

$$APS(u, a) = \frac{|total_user_packets(u)|}{T(t) - T(t - 1)}$$

$$total_user_packets = \{pk | pk \in pkt,$$

$$SourceIP(pk) = u,$$

$$flowlabel(flow(pk)) = a,$$

$$T - 1 \leq Timestamp(pk) \leq T\}$$

- Clustering Users: If a DDoS attack is detected, we cluster each source IP profile using DB-Scan algorithm. DB-Scan is an unsupervised algorithm that unlike other unsupervised clustering algorithms, doesn't required to specify the number of groups as hyperparameter. The algorithm operates on two key parameters: ϵ (epsilon), which defines the maximum

distance for two points to be considered neighbors, and $MinPts$, which is the minimum number of points required to form a dense region (a core point) [33], [34]. We set ϵ as 0.5 and $MinPts$ as 5. The hyperparameters of DBSCAN, primarily depend on the structure and density distribution of the data. When the ϵ and $MinPts$ values are set too low or too high, the clustering effectiveness of the model is significantly reduced, losing its core advantages [35]. As in modern DDoS attacks, there are a group of IPs where each of them act the same and share the same attack profile (zombies), we identify clusters that homogeneity in them is above a certain threshold as malicious clusters. **We can also block anomalous clusters as attackers.**

- Rule generation: After malicious users are detected, we generate the $\langle application, user \rangle$ which can be forwarded to a firewall to block the malicious traffic. This allows us to selectively block or limit the associated traffic, thereby preventing disruption to legitimate users and minimizing collateral damage.

The structure of ADD is depicted in Figure 5, which demonstrates the overall workflow of the system.

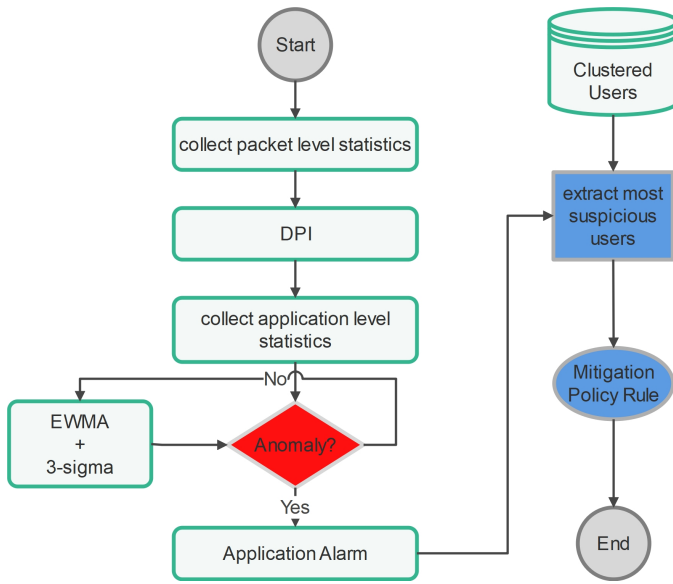


Fig. 5. Overall operation of ADD

4 EXPERIMENTS AND EVALUATION

5 DISCUSSION

6 CONCLUSION

7 APPENDIX

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